Gears and wheels

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use gear ratio and wheel circumference to predict an ideal travel speed.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 18.*
15 J; heating or sound are possible.

02. Worked example: explain the model
Speed ratio driven/driver=10/30=1/3; directions opposite for an external pair.

03. Read and interpret the model
Larger driven gear turns more slowly; meshed external gears turn opposite ways.

04. Apply the physics
2 rev/s; 0.40 m/s.

05. Explain your reasoning
Larger wheels need more axle torque for the same driving force, may add inertia/load, and may slow the motor or slip.

06. Review: connect an earlier idea *Review: Unit 7, Day 9.*
7 A in each branch.

07. Exit check
Doubles ideal speed; actual rpm/load/slip can change.

08. STRETCH: test the model
Examples 10: 30 and 15: 45; need motor speed, wheel size, slip and load.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 18.*
$|emf| = N|\Delta\Phi|/\Delta t = 0.5$ V.

02. Worked example: explain the model
Speed ratio driven/driver$=10/30=1/3$; directions opposite for an external pair.

03. Read and interpret the model
Larger driven gear turns more slowly; meshed external gears turn opposite ways.

04. Apply the physics
200 rpm; $C=0.1885$ m; speed$=0.628$ m/s.

05. Explain your reasoning
Larger wheels need more axle torque for the same driving force, may add inertia/load, and may slow the motor or slip.

06. Review: connect an earlier idea *Review: Unit 7, Day 9.*
Parallel 5 Ω; total 12 Ω; supply current 2 A.

07. Exit check
Torque ×3 and speed ×1/3, ignoring losses.

08. STRETCH: test the model
300 rpm$=5$ turns/s; $v = 2\pi(0.03)\times 5 \approx 0.942$ m/s.